

CS 112

Introduction to Data Structures

WEEK 0

Intro to C++

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This Week

- 1 Structure of a C++ program
- 2 Compilation pipeline
- 3 Types, variables, I/O
- 4 Functions
- 5 Pointers **KEY**
- 6 Arrays
- 7 C++ vs. Python

Your First C++ Program

```
1  int main() {  
2  
3  
4      return 0;  
5  }
```

Every C++ program execution begins at `main()`.

Your First C++ Program

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      cout << "Hello, World!" << endl;
6      return 0;
7  }
```

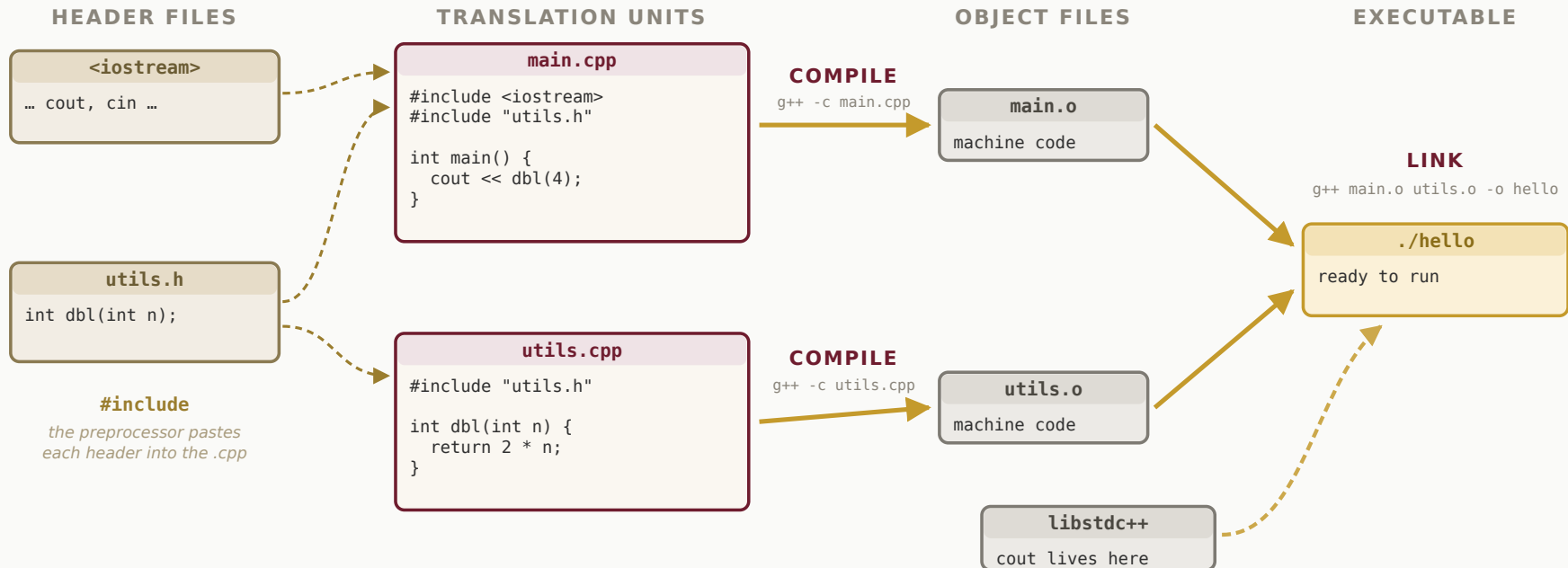
This is a full C++ program, with `#include`, `namespace`, and our `main()` function.

Anatomy of a C++ Program

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      cout << "Hello, World!" << endl;
6      return 0;
7  }
```

- 1 **#include <iostream>** Pulls in the I/O library, C++'s version of `import`.
- 2 **using namespace std;** Lets you write `cout` instead of `std::cout`.
- 5 **cout << ... << endl;** Prints to the terminal. `<<` is the “injection” operator. `endl` ends the line *and* flushes the buffer, so the text appears immediately.
- 6 **return 0;** Tells the OS the program finished cleanly.

Compilation Pipeline



COMPILE-TIME ERROR

error: 'cout' was not declared in this scope

One file didn't make sense on its own: a typo, a missing `#include`, a type mismatch.

LINK-TIME ERROR

undefined reference to `'helper()'`

Every file compiled fine, but a function was *declared* and never *defined*.

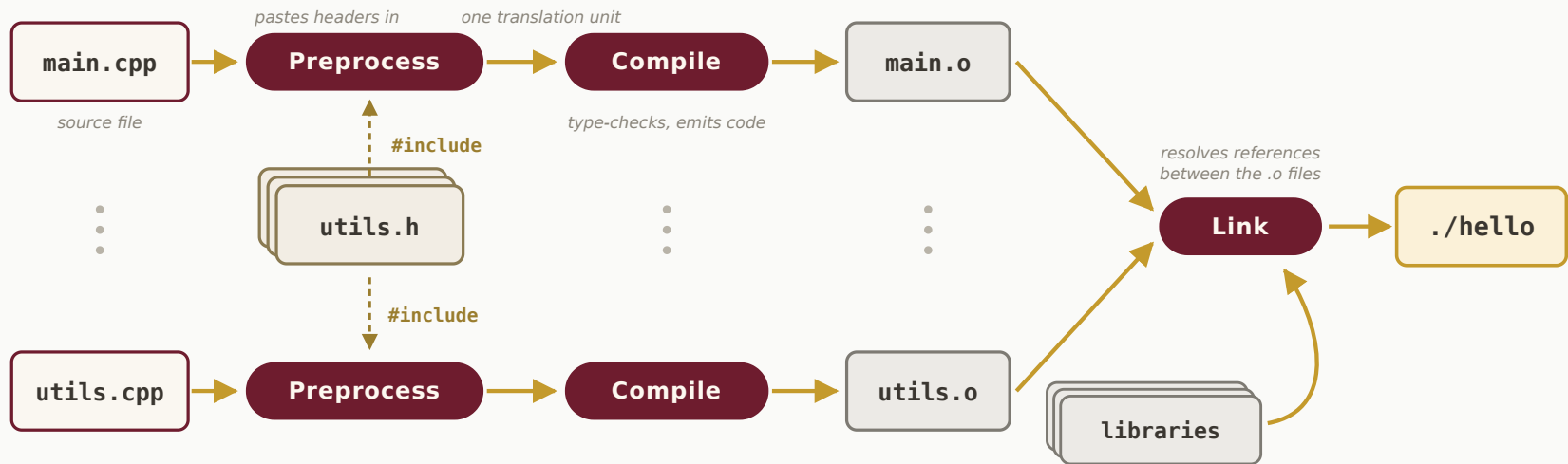
Headers and Translation Units

header .h

Declarations: what exists and how to call it. Shared by many `.cpp` files, and never compiled on its own.

translation unit .cpp

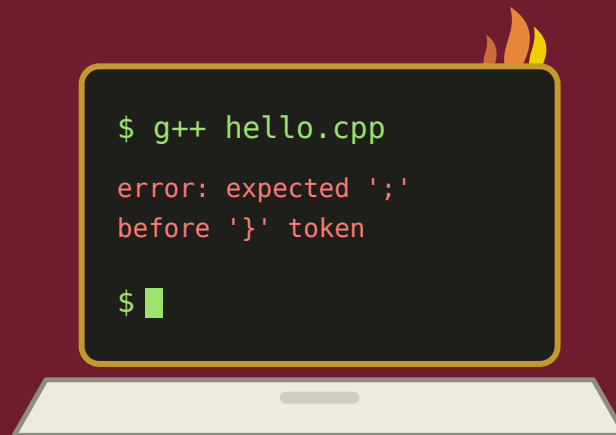
One `.cpp` after its headers have been pasted in. This is what the compiler actually reads, and it becomes exactly one `.o`.



Unlike Python, you *must* compile before running.

Upside: a whole class of bugs is caught at compile time, not in production.

Demo



Live coding. What could possibly go wrong?

(the semicolon is always missing on line 12)

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What does a compiler do?

- A.** Create an executable program from source code.
- B.** Check the source code for errors.
- C.** Pull in required libraries.
- D.** A and B only.
- E.** B and C only.
- F.** A, B, and C.
- G.** None of the above.

Data Types & Sizes

bytes	1	2	3	4	5	6	7	8		
bool									1 byte	<i>true / false</i>
char									1 byte	<i>'A', '\n'</i>
short									2 bytes	<i>-32 thousand to 32 thousand</i>
int									4 bytes	<i>-2 billion to 2 billion</i>
unsigned									4 bytes	<i>0 to 4 billion</i>
float									4 bytes	<i>about 7 digits of precision</i>
double									8 bytes	<i>about 15 digits of precision</i>
string									varies	<i>the text lives on the heap, so it can grow</i>

 whole numbers  real numbers  single values  grows as needed

```
cout << sizeof(int);      // 4: check any type yourself
cout << sizeof(double);   // 8
cout << sizeof(grade);    // 1: works on variables too
```

Types are **fixed at compile time**.

Sizes can vary by machine, `sizeof` tells you the truth on yours.

A `string` keeps its characters on the **heap**, memory the program asks for while it runs. More on that in week 4.

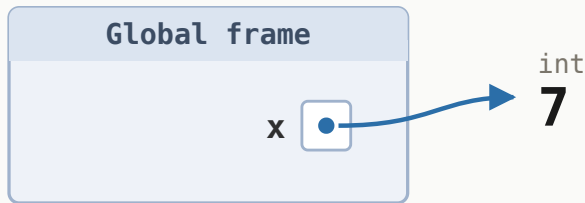
Python vs. C++: Variables

Python

```
x = 7
```

Frames

Objects



the name is a label; the object knows its own type

C++

```
int x = 7;
```

The stack

The heap



the box is on the stack, with a fixed type and address

Both look like "a variable called x holding 7".

In C++ that box lives on the **stack**, inside the *stack frame* belonging to this call of `main()`. Watch what happens when we assign `x = 3.14`.

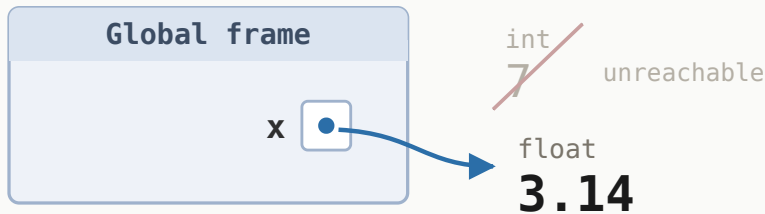
The Same Variable, Reassigned

Python

```
x = 3.14
```

Frames

Objects



same name, new object, new type

C++

```
x = 3.14;
```

The stack

The heap



same box on the stack: same address, same type

Python moves the label: `x` points at a *new* float object, and the `int 7` is abandoned.

C++ keeps the box: same address, still an `int`, so 3.14 is truncated to **3**.

Declare, Then Initialize

```
...  
int    count;           // declared: NOT initialized  
int    total  = 0;      // declared and initialized  
double pi    = 3.14159;  
char   grade  = 'A';  
bool   ready  = true;  
string name   = "Ada";  
  
cout << "count: " << count << endl;  // ⚠ undefined: could print anything  
cout << "total: " << total << endl;  // always 0  
...
```

```
$ ./init_demo  
count: -8593204  
total: 0  
$ ./init_demo  
count: 32764  
total: 0
```

*whatever those bytes happened to hold
exactly what you put there
same program, run again
different garbage this time*

In Python a name doesn't exist until you assign to it.

In C++ the box exists the moment you *declare* it, holding whatever the last program left in that memory.

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Will this C++ code compile?

```
int x = 11;  
x = "12";
```

- A. Yes
- B. No
- C. Yes, but with warnings.

Using `const`

```
const int    MAX_SIZE = 100;
const double PI       = 3.14159265;

MAX_SIZE = 200;  // ❌ compiler error: read-only!
```

- Put `const` before the type to make a variable read-only
- The compiler **rejects** any code that tries to change it
- Convention: use `ALL_CAPS` names for constants

Use `const` liberally; it documents intent and catches bugs at compile time.

Input / Output

```
#include <iostream>
using namespace std;

int main() {
    double value;
    cout << "Enter a number: ";    // output → terminal
    cin >> value;                  // input ← terminal
    cerr << "got " << value << endl; // error stream

    cout << "You entered: " << value << endl;
    return 0;
}
```

`cout` + `<<`: **injection**
output to the terminal

`cin` + `>>`: **extraction**
input from the terminal

Streams



A stream is a **one-way sequence of characters**, not a function call.

`cin` flows in, `cout` and `cerr` flow out.

Swap the keyboard for a file and the picture does not change: that is

`ifstream`, and it is the next slide.

File I/O

```
#include <iostream>
#include <fstream>      // ifstream reads a file, ofstream writes one
#include <cmath>         // sqrt
#include <cassert>
using namespace std;

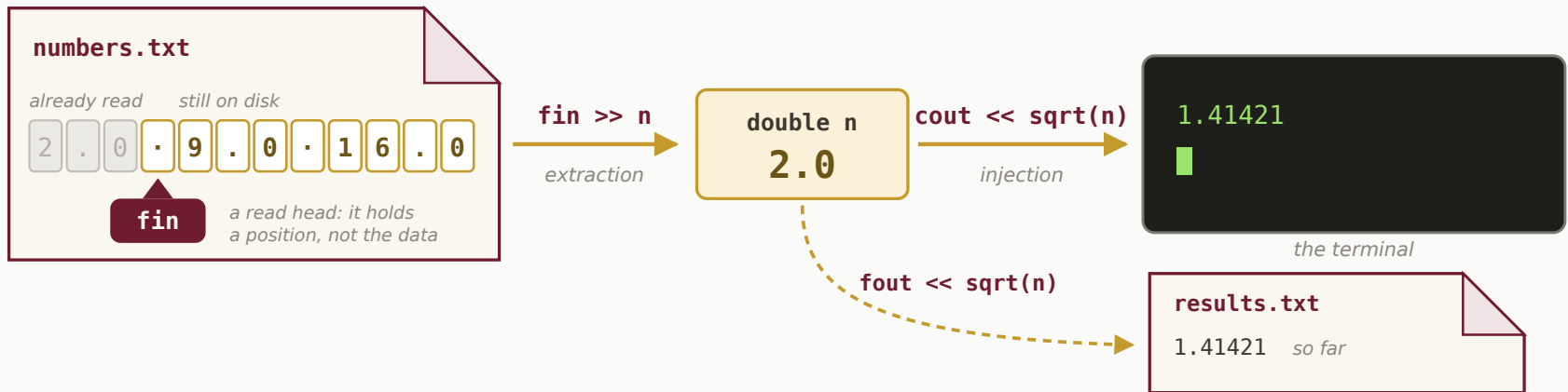
int main() {
    ifstream fin("numbers.txt"); // open the file, attach a stream to it
    assert(fin.is_open());       // a missing file is not an error to ignore

    double n;                    // ONE variable, reused for every number
    while (fin >> n) {           // read the next number; false when none left
        cout << sqrt(n) << endl; // fin has already moved past that number
    }

    fin.close();                 // hand the file back to the OS
    return 0;
}
```

`ifstream` reads from a file exactly like `cin` reads from the terminal. The loop ends on its own, because `fin >> n` is false once the file runs out.

What `fin >> n` Actually Does



The data never leaves the file: `fin` is a **read head** parked in it. Every `>>` converts the next characters into a `double` and drags that head forward. `<<` runs the other way, and does not care where it is pointed: `cout` for the screen, an `ofstream` for a file.

Decimal, Binary, Hexadecimal

```
int x = 33;           // decimal (default)
int y = 0x21;         // hex:      2×16 + 1 = 33
int z = 0b100001;     // binary:  1×32 + 1 = 33

cout << x << " " << y << " " << z << endl;
// Output:  33 33 33
```

- `0x` prefix → hexadecimal (base 16)
- `0b` prefix → binary (base 2)
- Memory addresses are typically displayed in hex

Three ways to write the *same number*. The compiler stores them identically in memory.

Meet Rocky



Not this Rocky.

Prof. Rocky Chang, our department



THIS Rocky!

Rocky, an Eridian, from Project Hail Mary

Why Ten?



*Three limbs down, two up: two hands of three fingers,
so he counts in sixes.*

how many things	0	1	2	3	4	5	6	7	8	9	10	11	12
we count in tens	0	1	2	3	4	5	6	7	8	9	10	11	12
Rocky counts in sixes	0	1	2	3	4	5	10	11	12	13	14	15	20
computers count in twos	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100

Ten is an accident of anatomy: we have ten fingers.

Rocky stands on three of his five limbs, leaving two hands of three fingers, so Eridians count in **sixes**: their "10" is our 6.

A wire is either on or off, so computers count in **twos**, and hex packs four of those bits into one digit.

Why bother with binary?



Everything on the machine is bits. Hex is just a friendlier way to read them.

xkcd #99 · xkcd.com · CC BY-NC 2.5

Functions

Write the tests *before* the function; it forces you to think about what it should do:

```
assert(factorial(0) == 1);  
assert(factorial(1) == 1);  
assert(factorial(4) == 24);  
assert(factorial(5) == 120);
```

This test-first style is called **test-driven development**. Your assertions become the spec.

Functions

```
unsigned factorial(unsigned n) {  
    unsigned result = 1;  
    for (unsigned i = 2; i <= n; ++i) {  
        result *= i;  
    }  
    return result;  
}  
  
int main() {  
    assert(factorial(0) == 1);  
    assert(factorial(1) == 1);  
    assert(factorial(4) == 24);  
    assert(factorial(5) == 120);  
    cout << "All tests passed!" << endl;  
    return 0;  
}
```

Pointers

Memory is a row of numbered boxes. A variable is one box.

```
int y = 77;
```

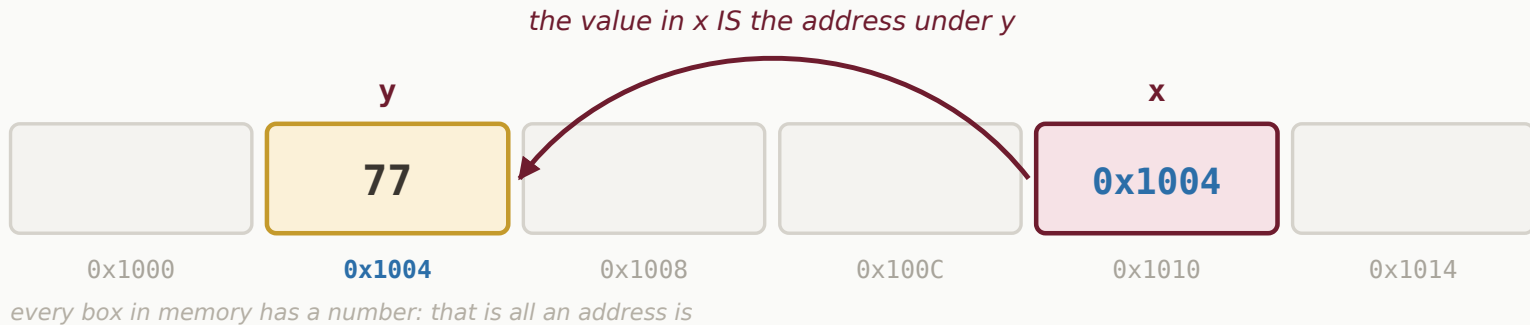


`y` is the box at `0x1004`. The name is ours; the machine only knows the number.

Pointers

`&` ("address-of") hands you the number of a box.

```
int y = 77;  
int *x = &y; // x holds the address of y
```



`x` is an ordinary box too. What makes it a pointer is *what it holds*: an address.

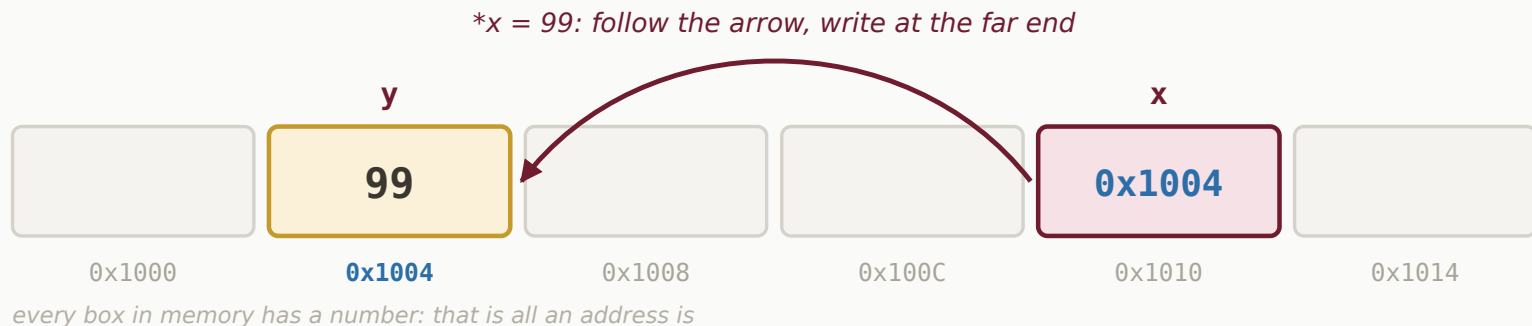
The arrow is us drawing the same number twice. The machine just has `0x1004`.

Pointers: Dereferencing

* means "go to that address".

```
int y = 77;
int *x = &y;

cout << x;    // 0x1004: the address itself
cout << *x;   // 77: follow the arrow
*x = 99;      // write at the far end of the arrow
cout << y;    // 99: y changed, without naming y
```



x and ***x** are different questions: *which box* versus *what is in it*.

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```
int y = 77;    // placed at address 0x1004
int *x = &y;   // placed at address 0x1010
cout << x;
```

What is printed?

- A. 77
- B. 0x1004
- C. 0x1010
- D. Error

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```
int y = 77;  
int *x = &y;  
cout << *y;    // note: *y, not *x
```

What is printed?

- A. 77
- B. 0x12345678
- C. 0x1010
- D. Compile error

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```
int y = 77;  
int *x = &y;  
cout << *x;
```

What is printed?

- A. 77
- B. 0x12345678
- C. 0x1010
- D. Error

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```
int y = 77;  
int *x = &y;  
*x = 78;  
cout << *x;
```

What is printed?

A. 77

B. 78

C. 0x1010

D. Error

TALK TO YOUR NEIGHBOR · TTYN

```
int y = 77;  
int *x = &y;  
*x = 78;  
cout << y;    // note: y, not *x
```

What is printed?

A. 77

B. 78

C. 0x1010

D. Error

Ask a programmer for a few pointers



He did give him pointers. Three of them, and every one is a real address.

xkcd #138 · xkcd.com · CC BY-NC 2.5

Arrays

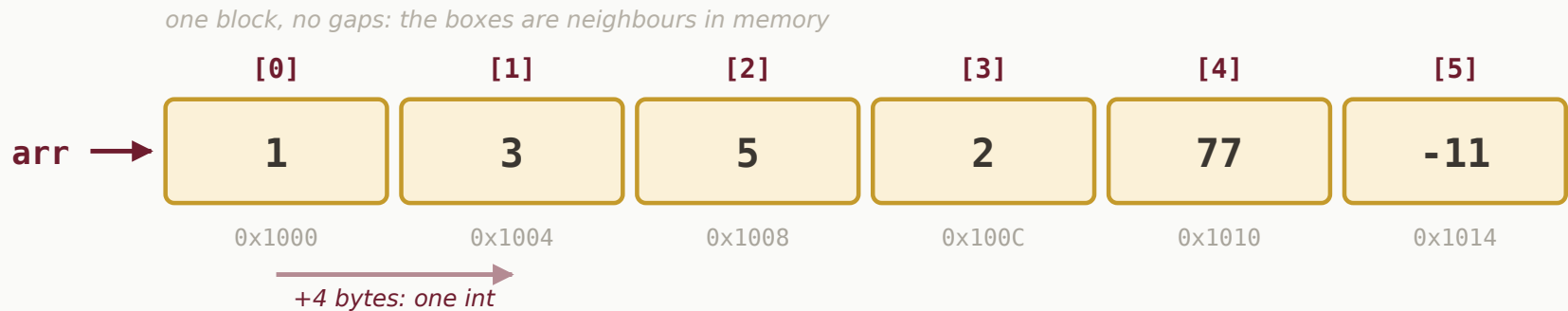
Like Python lists, but **fixed size** and **no bounds checking**.

```
char letters[26];           // 26 chars, uninitialized
float scores[10] = { 0 };   // all 0.0
int vals[] = { 10, 9, 4, 2, 0 }; // size inferred: 5
```

- Size must be known at compile time (for stack arrays)
- Indexes: `0` to `n-1`
- Accessing out of bounds → **undefined behavior**, no crash guaranteed

Arrays: Memory Layout

```
int arr[6] = { 1, 3, 5, 2, 77, -11 };
```



`arr` is a pointer to the first box, so `arr[i]` means "start at `arr`, step `i` ints along".

That is why indexes start at **0**: the first element is zero steps from the start.

Pointers & Arrays

```
int arr[] = { 3, -11, 4, 12 };  
int *ptr = &arr[2];           // ptr points at arr[2]  
  
cout << *ptr << endl;         // this box  
cout << *(ptr + 1) << endl;    // one int further along  
cout << *(ptr - 1) << endl;    // one int back
```

```
$ ./ptr_demo
```

```
4
```

```
12
```

```
-11
```

*arr[2], where ptr points
arr[3], the next box
arr[1], the box before*

`ptr + 1` does not add 1 to the address: it adds **one int**, so 4 bytes.

That is exactly what `arr[i]` has been doing all along.

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```
int arr[] = { 3, -11, 4, 12 };  
int *ptr = &arr[2];  
cout << *ptr;
```

What is printed?

A. 3

B. 2

C. 4

D. -11

C++ vs. Python: Core Differences

Feature	Python	C++
Variables	Reference to a typed object	Named memory location with explicit type
Type checking	Dynamic (runtime)	Static (compile time)
Function return type	Implied from <code>return</code>	Declared explicitly
1D data structure	<code>list</code>	<code>array</code> or <code>vector</code>
Memory management	Automatic (garbage collected)	Mostly explicit
Entry point	First line executed	<code>main()</code>

C++ vs. Python: More Differences

Feature	Python	C++
Interface / impl split	Not explicit	<code>.h</code> header + <code>.cpp</code> source
Parameter passing	Pass-by-value (objects by ref)	By-value, by-reference, by-const-ref
Access control	Mostly none	<code>public</code> / <code>private</code>
Constructors	Called explicitly	Explicit <i>or implicit</i> (scope entry)
Destructors	None (GC handles it)	Defined; called implicitly at scope exit
Generics	None	Templates, powerful but complex
Strings	Built-in	<code>#include <string></code>

Week 0 Recap

- C++ programs need a `main()` and must be **compiled** before running
- Types are **explicit and checked at compile time**
- Pointers hold **memory addresses**: the key concept in C++
- Arrays are contiguous memory blocks: fast, but no safety net
- C++ gives you more control than Python, and more responsibility

Next: Getting Started (pre-lab) · then a01: RPG Character Creator →

This deck stands on earlier CS112 materials by
Joel Adams and **Victor Norman** · adapted and extended by **Eric Araújo**

